

PPS 9.2: Resistivity, Thermistors, Electrical Power

1. Wire P has length L , diameter d , and a resistance of $1.00\ \Omega$. Another wire Q made from the same metal has length $3L$ and diameter $2d$. What is the resistance of wire Q ?

Resistance of wire P :

$$R_P = \frac{\rho L}{A}$$

$$A_P = \pi \frac{d^2}{4}$$

$$R_P = \frac{4\rho L}{\pi d^2}$$

Resistance of wire Q :

$$R_Q = \frac{\rho 3L}{A}$$

$$A_Q = \pi d^2$$

$$R_Q = \frac{3\rho L}{\pi d^2}$$

Then we can see that $4R_P = 3R_Q$ and thus $R_Q = \frac{4}{3}\Omega = 1.33\Omega$ (C)

2. The resistance of a wire of length L is 3.00Ω . The wire is extended so that its length becomes $1.50L$. Its volume remains the same. What is the resistance of the extended wire?

$$R_1 = \frac{\rho L}{A} = 3.00\Omega$$

If volume remains constant, and $V_1 = LA$ then $V_2 = \frac{3}{2}L \times \frac{2}{3}A = V_1$. Since $R_1 = \frac{\rho L}{A}$, then

$$R_2 = \frac{\rho 3^2 L}{2^2 A}$$

$$= \frac{9}{4}R_1$$

So the new resistance is $\frac{9}{4} \times 3 = 6.75$ (D).

3. A battery of e.m.f. of 8.0 V and internal resistance 2.5Ω is connected to an external resistor. The current in the resistor is 350 mA . What is the power dissipated in the external resistor?

$$P = IV = I^2 R = 0.350^2 \times 2.5 = 0.31\text{W} \text{ (D)}$$

4. A student determines the power P dissipated in a resistor. The measured values of the current I in the resistor and the resistance R of the resistor are:

$$I = (4.0 \pm 0.2)A \text{ and } R = (3.0 \pm 0.3)\Omega.$$

The equation $P = I^2 R$ is used to calculate P . What is the percentage uncertainty in the value of P ?

$$U_I = 0.2 = 5\%$$

$$U_R = 0.3 = 10\%$$

$$P = I \times I \times R$$

$$\text{so } U_P = U_I + U_I + U_R$$

$$= 20\% \text{ (B)}$$

5. A resistor of resistance R is connected in parallel with a resistor of resistance $2R$. The combination of resistors is connected to a cell.

What is the ratio $\frac{\text{power dissipated in resistor of resistance } R}{\text{power dissipated in resistor of resistance } 2R}$?

Let resistor A be the resistor with resistance R , and let resistor B be the resistor with resistance $2R$.

The resistors are connected in parallel, so the ratio of voltage drop is equal to the ratio of resistances.

So

$$P_A = \frac{V^2}{R}$$

$$P_B = \frac{(2V)^2}{2R} = 2 \left(\frac{V^2}{R} \right) = 2(P_A)$$

And therefore

$$\frac{\text{power dissipated in resistor of resistance } R}{\text{power dissipated in resistor of resistance } 2R} = \frac{1}{2} \text{ (B)}$$

6. **Two filament lamps X and Y are connected in series with a 16 V d.c. supply. The supply has negligible internal resistance.**

Lamp X emits a power of 2.0 W and lamp Y emits a power of 6.0 W. What is the potential difference across the lamp X ?

The current in the circuit is the same because of Kirchhoff's First Law. Then,

$$P_X = IV_X$$

$$P_Y = IV_Y$$

Since we know both P_X and P_Y , we can say that

$$2 = IV_X$$

$$6 = IV_Y$$

Then

$$\frac{6}{2} = \frac{V_Y}{V_X}$$

$$\frac{V_Y}{V_X} = \frac{3}{1}$$

Since we know that the d.c. supply is 16V and therefore the total p.d. across both filament bulbs must be 16V, we find that the p.d. in lamp X is 4.0V. (B)

7. **A household is planning to change all their 60 W filament bulbs to 12 W LED bulbs. The household has a total of 10 bulbs. Each bulb will be used for about 2000 hours in one year. The cost of 1 kWh is 15.4p. What would the annual saving be?**

For the filament bulbs, in 1 year:

$$\begin{aligned} E &= Pt \\ &= 60 \times (2000 \times 3600) \\ &= 432 \text{ MJ for one bulb} \end{aligned}$$

So for 10 bulbs, the energy used per year is 4320 MJ. We know that 3.6 MJ = 1 kWh, so 4320 MJ = 1200 kWh = £184.80 annually.

For the LED bulbs, in 1 year:

$$\begin{aligned} E &= Pt \\ &= 12 \times (2000 \times 3600) \\ &= 86.4 \text{ MJ for one bulb} \end{aligned}$$

So for 10 bulbs, the energy used per year is 864 MJ. We know that 3.6 MJ = 1 kWh, so 864 MJ = 240 kWh = £36.96 annually.

Therefore, the annual savings is £147.84 (C).

8. a. i. $P = IV = 30 \times 10^{-3} \times 10 = 0.3\text{W}$
 ii. The component is a filament lamp. As the current increases, the voltage increases exponentially. This is because the wire inside the lamp gets hotter, so resistance increases (because $R \propto T$), and voltage increases because it takes more p.d. to move the same amount of charge through the bulb.
 b. The resistance of resistor **X** can be found by

$$R_X = \frac{\rho L}{A}$$

The total resistance in the circuit is given as $R_X + R_Y$, and the current in the circuit can be found by rearranging $R = \frac{V}{I}$ to get $I = \frac{V}{R}$

$$\begin{aligned} I &= \frac{V}{R} \\ &= \frac{V}{R_X + R_Y} \\ &= \frac{V}{\frac{\rho L}{A} + R_Y} \end{aligned}$$

Then, by substituting in the given values, we get

$$I = \frac{3.0}{\frac{(1.5 \times 10^{-2})(8.0 \times 10^{-3})}{1.2 \times 10^{-6}} + 68} = 0.018\text{A}$$

9. a. By using a micrometer to measure the diameter of the wire, and measuring at several points along the wire to increase the precision of the measurement.
 b. i. $\rho = \frac{RA}{L} = \frac{1.86 \times 17 \times \pi \times (6 \times 10^{-5})^2}{21} = 1.703 \times 10^{-8} \Omega\text{m}$
 ii. The uncertainties of the diameter, resistance, and length are as follows:

$$U_d = 0.01 = 8.33\%$$

$$U_R = 0.02 = 1.08\%$$

$$U_L = 0.1 = 0.476\%$$

Since $\rho = \frac{RA}{L}$,

$$A = \pi \frac{d^2}{4}$$

$$U_A = 2(U_d) = 16.7\%$$

$$\therefore U_\rho = U_A + U_R + U_L = 18\%$$

10. a. The resistance can be calculated by $R = \frac{V}{I}$. Then

$$\begin{aligned}\rho &= \frac{RA}{L} \\ &= \frac{VA}{IL} \\ &= \frac{1.80 \times (1.3 \times 10^{-7})}{0.26 \times 0.75} \\ &= 1.2 \times 10^{-6} \Omega \text{m}\end{aligned}$$

- b. The cross-sectional area is very small, so the percentage uncertainty of the measurement is most likely quite large. If the wire had a larger cross-sectional area, it would reduce this uncertainty, and the measurement would become more accurate. Therefore, the predicted resistivity would be closer to the true value.

Extension Questions

Bulbs and Power

Let the resistor in series be resistor s , and the resistor in parallel be resistor p .

The power in the bulb is given by

$$P_b = \frac{V_b^2}{R_b}$$

Then the voltage in the bulb is given by

$$V_b = \sqrt{P_b R_b}$$

Let V be the total voltage in the system. Therefore

$$V = V_s + V_b$$

$$V_s = V - V_b$$

We can also say that

$$I = I_p + I_b$$

$$\frac{V_s}{R} = \frac{V_p}{R} + \frac{V_b}{R_b}$$

And since resistor p and the bulb are in parallel, their voltages are the same: $V_p = V_b$.

So

$$\frac{V_s - V_b}{R} = \frac{V_b}{R_b}$$

And therefore

$$\begin{aligned} R &= \frac{R_b(V_s - V_b)}{V_b} \\ &= \frac{R_b(V - 2V_b)}{V_b} \\ &= \frac{R_b(V - 2\sqrt{P_b R_b})}{\sqrt{P_b R_b}} \end{aligned}$$

By substituting in values:

$$\begin{aligned} R &= \frac{10(230 - 2\sqrt{100 \times 10})}{\sqrt{100 \times 10}} \\ &= 52.8\Omega \end{aligned}$$

Melting a Snowman

- a) $P = IV = 6 \times 0.115 = 0.69\text{W}$. $0.69 \times 2 = 1.38\text{W}$
 b) $m = \left(\frac{4}{3}\pi \times 0.5^3 + \frac{4}{3}\pi \times 1^3\right) \times 930 = 4400 \text{ kg}$
 c) The LEDs are 30% efficient, so they emit $1.38 \times 0.7 = 0.966$ joules per second.

The energy required to melt ice is $335 \text{ J g}^{-1} = 3.35 \times 10^5 \text{ J kg}^{-1}$.

Therefore for 4400kg it takes $3.35 \times 10^5 \times 4400 = 1.474 \text{ GJ}$ of energy to melt the snowman.

At 0.966 J/s, the snowman will take $\frac{1.474 \times 10^9}{0.966} = 1,525,879,917$ seconds or 48.4 years to melt fully.

Heating a wire

We know that $R \propto T$. We also know that $P \propto \Delta T$ because the wire maintains a steady temperature. Therefore

$$I^2 R \propto \Delta T$$

$$I^2 T \propto (\Delta T)$$

$$\therefore I^2 T = k \Delta T$$

Let $T_1 = 330^\circ\text{K}$ and $T_2 = 660^\circ\text{K}$, and $T_a = 300^\circ\text{K}$ = the temperature of the stream of air. Let I_1 be the current needed to maintain a steady temperature of $57^\circ\text{C} = 5.00\text{A}$, and let I_2 be the current needed to maintain a steady temperature of 387°C . Then

$$\begin{cases} I_1^2 T_1 = k(T_1 - T_a) \\ I_2^2 T_2 = k(T_2 - T_a) \end{cases}$$

By dividing the bottom equation by the top, we get

$$\frac{I_2^2 T_2}{I_1^2 T_1} = \frac{T_2 - T_a}{T_1 - T_a}$$

$$I_2 = I_1 \sqrt{\frac{T_1(T_2 - T_a)}{T_2(T_1 - T_a)}}$$

$$= 5 \sqrt{\frac{330(660 - 300)}{660(330 - 300)}} = 5\sqrt{6}$$

$$= 12.2\text{A}$$